lets them use the testing environment they need at the moment and pay only for that environment. Users can have all the software and hardware they might need at their disposal, while only paying for it when they actually use it.

Cloud-based testing avoids any errors made during environment configuration and their repetition across all devices. Software testing in the cloud is available to testers at anytime, anywhere. Cloud-based testing allows software companies to better implement DevOps, which requires collaboration between developers and testers.

Types of cloud testing

The following order may be helpful for testers in getting ready to use cloud-based testing tools:

1. Set clear objectives.
2. Create testing strategy.
3. Plan the infrastructure.
4. Select a reliable provider.
5. Determine service access.
6. Use free trials.
7. Monitor and analyse test results.

In a cloud environment, any application can be subjected to three types of testing: functional, non-functional and ability.

Functional testing: This ensures that the software meets functional requirements, which can be of three types:

- System testing that analyses the system's behaviour and design, and how it meets the customer's expectations.
- Acceptance testing that ensures whether the application meets certain needs of its users.
- Integration testing that proves that the application is compatible with different platforms and works well while moving from one cloud infrastructure to another.

Non-functional testing: This helps to check the non-functional aspects of software such as its performance, usability and reliability. This can be done at three levels:

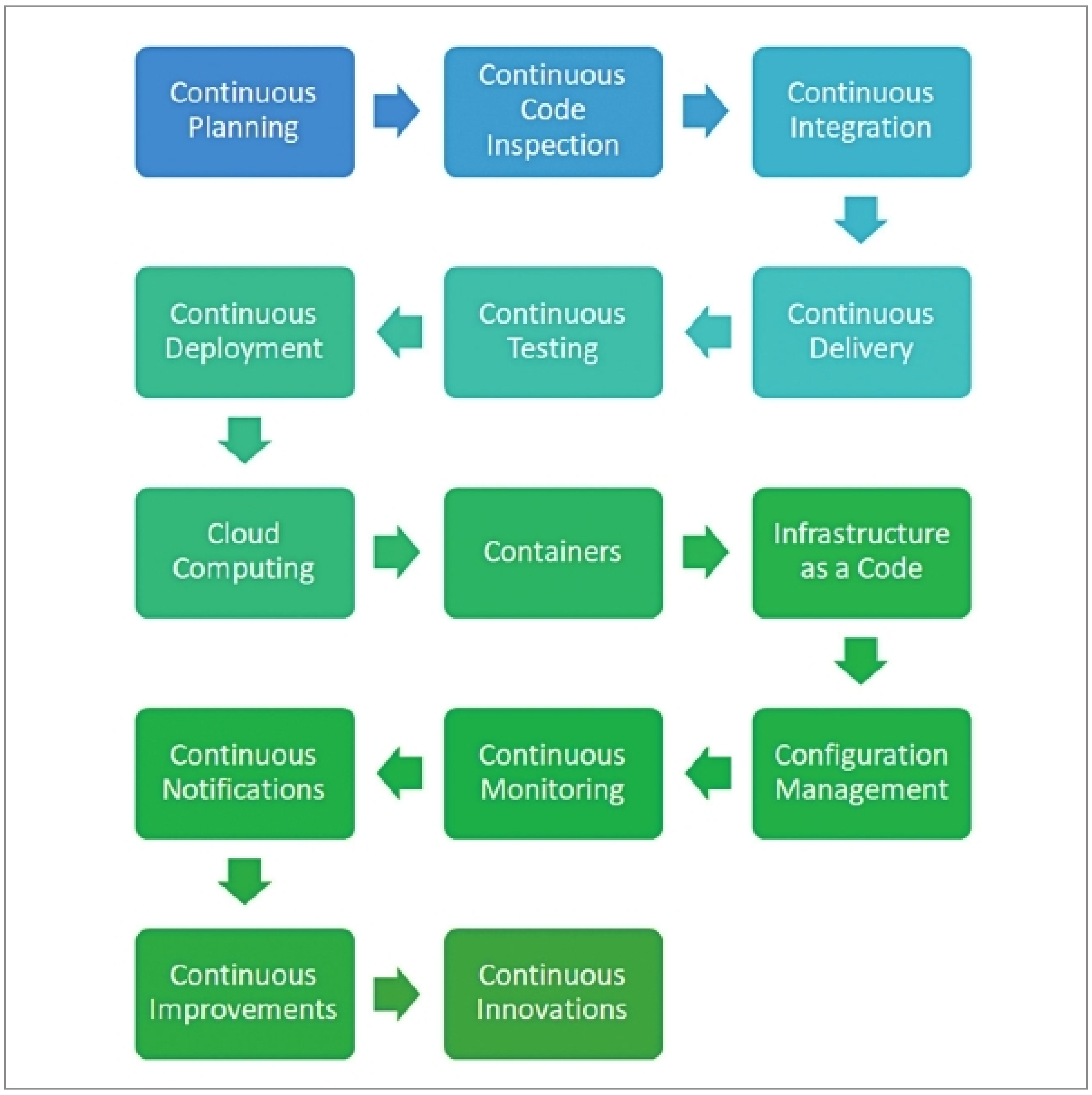

Figure 1: Continuous practices

- Business requirement testing that verifies how precisely an application meets the specified business requirements.
- Security testing to ensure that data is stored and transmitted safely.
- Scalability and performance testing to measure software response time while the system is subjected to increasing load.

Ability testing: This verifies whether users really receive application services on demand. To check this, the test team can conduct the following types of testing:

- Compatibility and interoperability testing to evaluate the application's compatibility with various environments and platforms.
- Disaster recovery testing to evaluate disaster recovery time and ensure that the application becomes available to users again with minimum data loss.
- Multi-tenancy testing to verify whether the application can ensure a sufficient level of security and access control when multiple users invoke it in the cloud.

Challenges of cloud-based testing

Although cloud-based testing has many benefits for assuring quality, it also has operational challenges that the testers should be ready to overcome. Vendors of cloud-based tools provide terms and conditions of their cloud services that differentiate the responsibilities of the vendor and the cloud user.

There is a lack of standards for how testers can integrate internal resources of their companies' data centres with public cloud resources. Since architectures and operating models are developed by public cloud providers, these cloud services have little interoperability. Thus, testers may face compatibility issues if they want to switch cloud vendors because different cloud vendors might not offer the same services.